The intensive cultivators
Karat 10
Karat 12

LEMKEN THE AGROVISION COMPANY

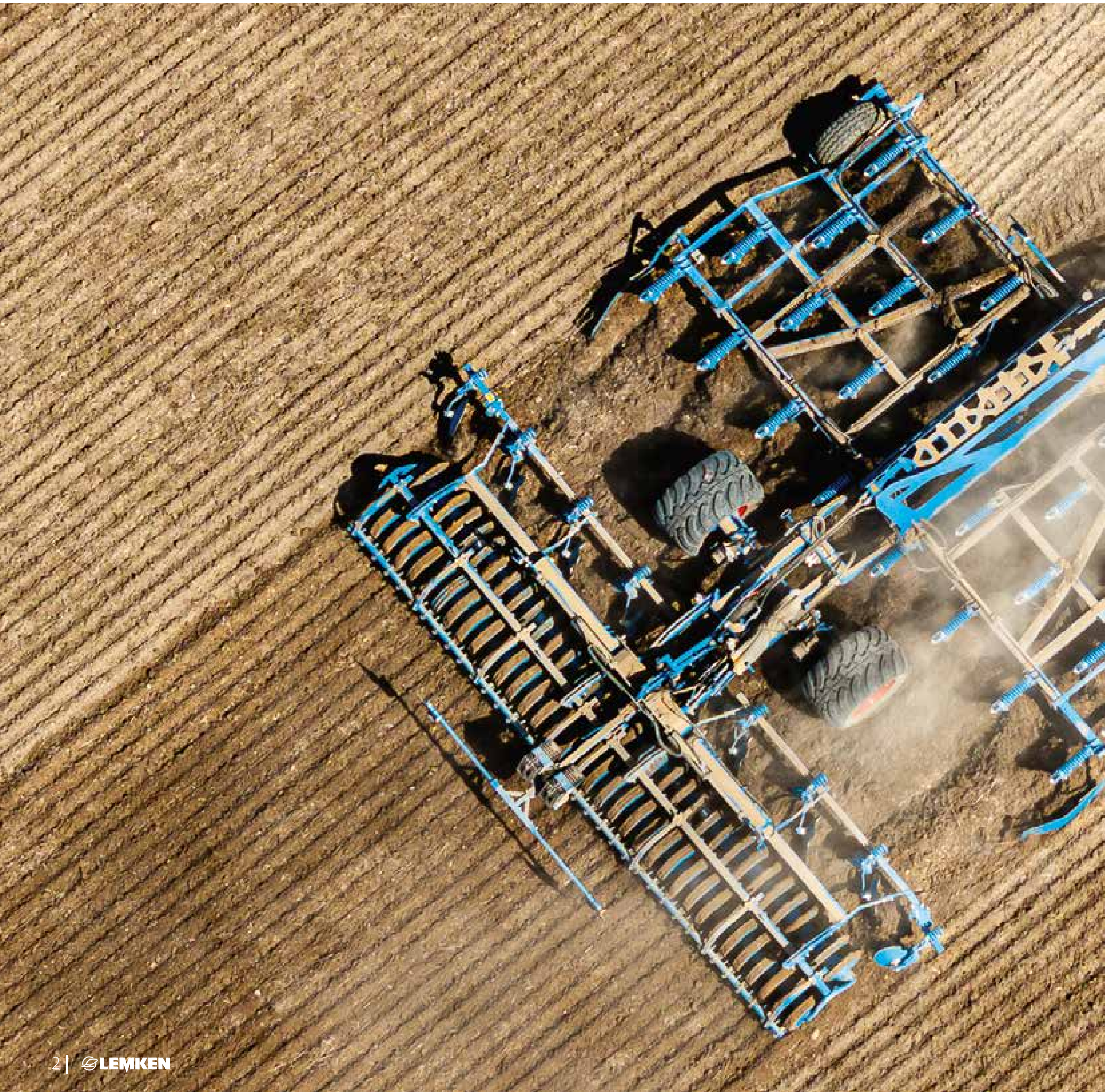
Every detail a highlight

Share shapes and variants

Levelling, loosening or crumbling. A variety of share, ranging from narrow to wide, are available for flexible use. This allows the intensity of tillage to be optimally adapted to the conditions and requirements at hand.

Symmetry

Thanks to the symmetrical arrangement of the tines relative to the drawbar axis, the Karat works without side draught. The (optional) depth control wheels positioned within the working width ensure that the cultivator is optimally guided. This improves the Karat's stability and directional stability, which has an extremely positive effect on the tractive power requirement.



Frame design

The more beams there are, the smaller the line spacing can be. Combined with a generous 80 cm underframe clearance, this ensures blockage-free operation with both the 3-beam Karat 10 and the 4-beam Karat 12.

Trailing units

The right trailing unit can be selected from an extensive range of rollers to achieve optimum reconsolidation and crumbling effects. An additional single-row harrow creates optimal germination conditions for volunteer cereals and weed seeds and removes soil from roots.

The perfect frame for every use

The line spacing affects the intensity of tillage, while the number of beams determines the evenness of tillage. The choice between a 3- or 4-beam cultivator, a Karat 10 or 12, should therefore be based on the specific soil conditions and the desired tillage outcomes. As a general rule, more beams achieve better soil loosening and mixing; however, the power requirement and the length of the implement usually increase as well.

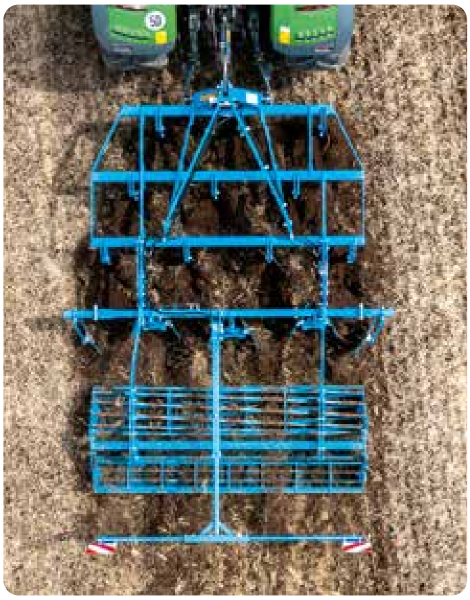
With more beams, the tines are arranged in several rows, which increases the line distance between the tines on adjacent beams and reduces the risk of blockages that may be caused by high crop residues.

There is a choice between a short and a long drawbar for attaching the cultivator to the tractor. This allows for an external tractor width of 3.2 m or 4.5 metres. The running gear of the semi-mounted Karat is integrated into the cultivator, making it extremely compact, stable and highly manoeuvrable. This position ensures smooth soil flow and optimal levelling.

Thanks to the favourable weight distribution, heavy trailing rollers can be used for thorough reconsolidation. If the Karat is not fitted with an optional trailing roller, depth control is handled by the integrated running gear.

Karat 10

The new Karat 10 reveals its efficient design even at first glance. The tines are now arranged **symmetrically relative to the drawbar axis**, ensuring that the machine works without side draught. This makes the 3-beam cultivator extremely easy to pull and guarantees intensive mixing at the same time.



The 3-beam Karat 10 cultivator is particularly well suited for incorporating crop residues and for shallower tillage. The **narrower line spacing** of 30 cm for the mounted and 31 cm for the folding mounted and semi-mounted versions enables intensive tillage.

The semi-mounted Karat 10 can optionally be fitted with a leading disc section. This ensures that large volumes of organic matter can be incorporated even better. The discs have a diameter of 520 mm and are individually suspended on leaf springs. They chop organic matter and break up coarse soil clumps so that the cultivator is able to mix soil intensively and work **without blockages**.
The discs are convenient to adjust hydraulically to the desired working depth from the tractor seat.

Karat 12

The 4-beam Karat 12 is the ideal choice for larger volumes of crop residues or if **very intensive tillage** is required, thanks to its large line distance.

The Karat 12 cultivator is particularly well suited for heavy soils and no-tillage cultivation methods for both medium-deep and deep tillage. With a line spacing of 90–80–90 cm, **four rows of tines** and a line distance of 23.5 cm, there is sufficient clearance for blockage-free operation.

The symmetrical tine arrangement effectively minimises side draught, prevents unilateral dipping of the implement into the soil and optimises pass alignment. The position of the **standard depth control wheels** next to the first row of tines and within the working width ensures that the cultivator is optimally guided.

Shallow to deep – and even intensive if required

Stubble tillage is usually carried out in several stages, and the preferred share shape depends on the desired outcome. The choice of share shape should be adapted to soil conditions, crop residue type and desired tillage outcome. For the first, shallow tillage pass, wing or duck-foot shares (DeltaCut shares) are often used to distribute crop residues evenly, control weeds and break capillary action. The goal of subsequent passes is then to loosen the soil more deeply.

From wing shares to narrow shares

Wing shares ensure that the soil is cut across the full width, even with a wide line spacing. If the Karat is to be used for (ultra) shallow first stubble tillage at working depths of around 5 cm, it can be fitted with DeltaCut shares. For subsequent passes with deeper tillage, share points are available in different widths, including special narrow shares.

Quick-change system

The Karat features a quick-change system as standard. The entire interchangeable share foot with wing shares can, for example, be easily removed using just a linchpin, without tools, and replaced by an interchangeable share foot with narrow shares for deep loosening. Worn parts can be replaced in the workshop without additional effort, e.g. for protecting the implement. This significantly reduces set-up times.

A tough performer

Carbide plate-reinforced shares are used for stubble tillage in situations where particularly high wear resistance is required. Typical areas of application are:

1. Dry, abrasive soils: in sandy or gravelly soils, conventional shares can wear very quickly. Carbide plates extend the service life considerably.
2. Intensive and deeper tillage: carbide ensures a longer service life for the shares.
3. Sharp cut: with DeltaCut shares, carbide helps to preserve the cutting edge, even when travelling at speed and working in dry and hard soils.

Soldered-on carbide plates in LEMKEN machines ensure an **extremely long service life** for share points. In addition, a reinforced carrier material delivers an optimal balance between carbide and steel wear. The targeted use of materials on the cones and overlap sections of the share points protects the attachment screw against wear.

A perfect finish with trailing units

To achieve optimum reconsolidation and crumbling effects, the right trailing unit can be selected from an **extensive range of rollers**. An additional single-row harrow creates optimal germination conditions for volunteer cereals and weed seeds and removes soil from roots.

If reconsolidation is not desired, the Karat cultivators can also be used without a roller. In this case, depth control is handled via the running gear. The semi-mounted Karat can be fitted with additional wheelmark eradicator tines to loosen wheel tracks.

The semi-mounted versions of the Karat intensive cultivator come with a **hydraulic working depth adjustment as standard**. The rigid and folding mounted cultivators come with mechanical working depth adjustment using movable bolts as standard.

As always, LEMKEN delivers **optimal operating quality**:
The infinitely variable hydraulic working depth adjustment can be operated from the tractor cab while driving, usually even without needing to readjust the levelling unit. The levelling unit may need to be adjusted, however, if the working depth is changed significantly. If the Karat is fitted with a roller, depth control is handled via the roller and the levelling unit can be adjusted separately.

Added value on top

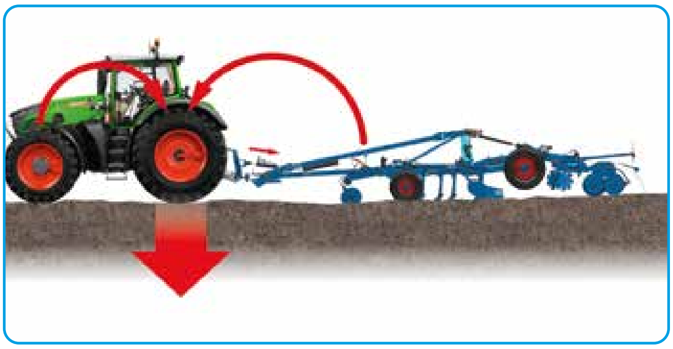
You want things level

For even levelling behind the tine section, you have the choice between two easily adjustable levelling elements that are integrated into the cultivator: **levelling tines** are fitted as standard, while **levelling discs** are optionally available for the Karat 10. Cultivators should produce level results. Optional boundary levelling tools, such as side shields or boundary levelling discs or tines, ensure optimal levelling along the outer edges of the cultivator. This supports precise pass alignment without ridging. Boundary levelling discs or tines can be folded up either mechanically or hydraulically, and the side shield can be pushed in. This makes it easy to work without boundary levelling tools, for example along fences. On the Karat 12 and the mounted Karat 10, the levelling tools can be adjusted individually using a bolt. The trailed Karat 10 models feature a centralised setting mechanism.

Unstoppable

The basic version features a shear bolt that protects the Karat if a tine becomes caught under a rock or root. Non-stop overload elements are available as an option for stony soils. These elements are additionally equipped with a shear bolt to protect the frame. At greater working depths, the large vertical deflection and high **trigger forces of 550 kg per tine** ensure reliable performance. The angle of the tines can be adjusted via the shear bolt to improve penetration in very dry or hard soils. Robust, welded beams and strong compression springs easily resist high continuous loads.

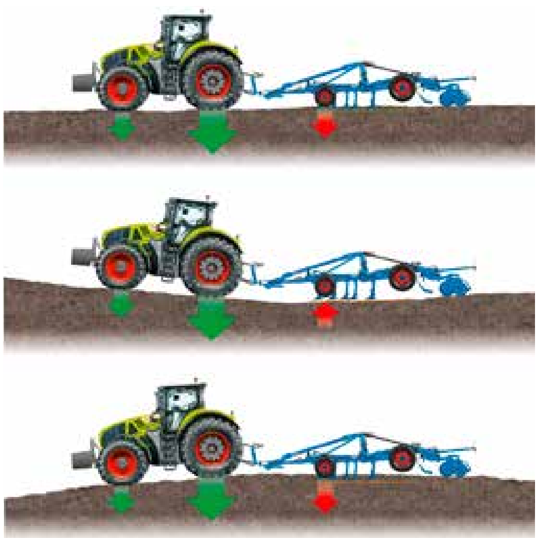
On track – optimal traction transmission



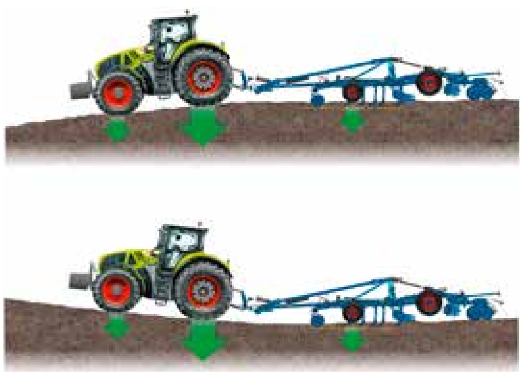
Traction booster

The semi-mounted versions of the Karat intensive cultivator feature an optional hydraulic traction booster. An additional hydraulic cylinder, which is preloaded via a gas pressure accumulator, is applied to the top link coupling point and the cultivator drawbar when the tractor and implement are coupled. At a pre-load pressure of **180 bar, an additional weight of 1.6 tonnes** applies to the rear tractor axle. The hydraulic traction booster therefore acts as a smart ballasting system:
the optimised distribution of forces reduces slip and therefore saves fuel. This ensures that tractors with less powerful output or less wheel ballast can also be used for cultivating heavy soils.

ContourTrack



Without ContourTrack



With ContourTrack

For working at a constant working depth in hilly terrain, the ContourTrack equipment option is recommended. This system uses the pivot point in the frame behind the working section and an additional hydraulic cylinder to prevent the machine from working too shallowly in depressions or too deeply on hills. The working depth is adjusted automatically.

ContourTrack therefore ensures that the tractor/ implement unit follows the ground contours optimally, providing the best possible results in hilly terrain.
This simple, yet effective system is available as an option for the semi-mounted Karat cultivators.

Always the right fit
Drawbars available in a range of types and lengths and with different coupling points allow the machine to be used with all tractors, including tractors with twin tyres and an overall width of up to 4.5 metres.

It’s in the connection

With iQblue connect, both new and existing cultivators can be equipped or retrofitted with intelligent hydraulic working depth adjustment. The combination of job computer, sensor kit and GPS technology significantly increases tillage efficiency.

When used with the cultivator, iQblue connect takes over precise, automated control of the working depth.
This enables the following:

- Exact implementation of application maps
- Automatic adjustment of the working depth to changing ground conditions.
- Centimetre-accurate display of the current working depth

Communication between the implement and tractor is handled via the ISOBUS TIM (Tractor Implement Management) standard, enabling iQblue connect to control a tractor’s hydraulic functions directly. As a result, the implement is actively integrated into the control process – the cultivator controls the tractor. As an added bonus, the job computer can be easily transferred to other machines in just a few simple steps. This makes the system highly flexible and cost-effective – ideal for a wide range of applications and farm sizes.

Specifications

Karat	Tines / Disc pairs (+ boundary discs)	Line distance [cm]	Beam distance [cm]	Working width [cm]	Length (max.) [cm]	Weight (without roller) [kg]	Power requirement (min.–max.) [kW/hp]
10/300 (U)	10/3	30	70/70	300	385	1,275	77/132 – 105/180
10/350 (U)	11/3	30	70/70	350	385	1,370	90/154 – 122/210
10/400 (U)	13/4	30	70/70	400	385	1,570	103/176 – 140/240
10/400 K(U)	13/4	30	70/70	400	415	1,880	103/176 – 140/240
10/500 K(U)	16/5	30	70/70	500	415	2,490	129/220 – 175/300
10/400 K(U)A	13/4	31	100/100	400	1,012*	4,200	103/206 – 140/280
10/500 K(U)A	16/5	31	100/100	500	1,012*	4,730	129/257 – 175/350
10/600 K(U)A	19/6	31	100/100	600	1,012*	5,120	154/304 – 245/420
10/700 K(U)A	22/7	31	100/100	700	1,012*	5,600	180/360 – 245/490
12/400 KUA	17/4	23.5	90/80/90	400	900	5,330	132/180 – 252/340
12/500 KUA	21/5	23.5	90/80/90	493	900	5,590	165/224 – 315/425
12/600 KUA	25/6	23.5	90/80/90	587	900	6,520	198/270 – 378/510
12/700 KUA	29/7	23.5	90/80/90	680	900	7,270	231/314 – 441/595

* with PPW furrow press profile roller, discs, leading unit and harrow

A WELL-ROUNDED SOLUTION.

At LEMKEN, we don't think in terms of isolated work steps – instead we look at the full cycle including all facets of agricultural engineering. The result is comprehensive solutions that intermesh perfectly. For you, this means: high-quality, future-oriented, efficient technology for an agriculture that is both profitable and sustainable.

LEMKEN GmbH & Co. KG
Weseler Strasse 5
46519 Alpen, Germany
Tel. +49 2802 81-0
Fax +49 2802 81-220
info@lemken.com

Find out more at
lemken.com

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